

Dataset Details

Name	Diabetes 130-US Hospitals for Years 1999-2008
Source	UCI Machine Learning Repository (donated 2014-05-02)
Link	https://archive.ics.uci.edu/dataset/296/diabetes+130-us+hospitals+for+years+1999-2008
DOI	10.24432/C5230J
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Citation	Strack, B., DeShazo, J., Gennings, C., Olmo, J., Ventura, S., Cios, K., & Clore, J. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 70,000 Clinical Database Patient Records. <i>BioMed Research International</i> , 2014.
Format	Two CSV files: <code>diabetic_data.csv</code> (101,766 rows, 50 columns) and <code>IDs_mapping.csv</code> (three lookup tables stacked in one file)
Purpose in this project	Teaching data for an end-to-end healthcare data system: ETL, quality control, a relational database, a secured REST API, KPI analytics and a dashboard. Educational use only.

1. What one row means

One row is one hospital **encounter** (an inpatient stay) of a diabetic patient at one of 130 US hospitals, 1999-2008. `encounter_id` is unique (101,766 distinct values). `patient_nbr` repeats: there are **71,518 distinct patients**, 16,773 of whom appear more than once (the most frequent has 40 encounters). Patients and encounters are therefore never used interchangeably in this project.

2. Key fields

Field	Meaning	How it is used
<code>encounter_id</code> , <code>patient_nbr</code>	Source keys (BIGINT, never generated)	Primary keys of <code>encounters</code> and <code>patients</code>
<code>age</code>	Ten brackets [0-10) ... [90-100)	Stored as <code>age_group</code> plus <code>age_order</code> 1-10; always sorted by <code>age_order</code>
<code>race</code> , <code>gender</code>	Demographics (sensitive)	Stored only on <code>patients</code> ; hidden from the analyst role; never a dashboard breakdown, never exported
<code>admission_type_id</code> , <code>admission_source_id</code> , <code>discharge_disposition_id</code>	Coded categories	Foreign keys to lookup tables loaded from <code>IDs_mapping.csv</code> (8, 25 and 30 rows)
<code>medical_specialty</code> , <code>payer_code</code>	Admitting specialty, payer	Lookup tables; NULL when missing
<code>time_in_hospital</code>	Length of stay in days (1-14)	Length-of-stay KPIs
<code>num_lab_procedures</code> , <code>num_procedures</code> , <code>num_medications</code> , <code>number_outpatient</code> , <code>number_emergency</code> , <code>number_inpatient</code> , <code>number_diagnoses</code>	Counts of services and prior visits	Stored on <code>encounters</code> ; prior inpatient visits drive the utilisation analysis

Field	Meaning	How it is used
diag_1, diag_2, diag_3	ICD-9 diagnosis codes (strings, include V and E codes)	Rows in encounter_diagnoses (position 1-3)
max_glu_serum, A1Cresult	Lab results; the text None means "test not performed"	Stored as ENUMs; None is a real value
23 drug columns (metformin ... metformin-pioglitazone)	No, Steady, Up, Down	Long table encounter_medications holding only Steady / Up / Down
change, diabetesMed	Medication change, diabetes medication prescribed	med_changed, diabetes_med booleans
readmitted	NO, >30, <30	The outcome; see section 4

3. Preprocessing performed (every rule)

3.1 Reading

- ? is read as missing (keep_default_na=False, na_values=['?', '']). The literal text None in max_glu_serum and A1Cresult is **kept**: with pandas defaults it would be turned into missing (96,420 and 84,748 rows), wrongly suggesting that most patients have no lab result.
- ICD-9 columns are read as text so that v and E codes and decimal points survive.

3.2 Columns removed

Column	Reason
weight	96.86% missing
examide, citoglipton	constant value (No) in every row

3.3 Transformations

- Column names to snake_case; change to med_changed (Ch = true), diabetesMed to diabetes_med (Yes = true).
- age to age_group plus age_order (1-10).
- diag_1..3 to three rows per encounter at most, in encounter_diagnoses.
- The 21 informative drug columns to long format; rows with No are **not stored** (120,054 prescribed cells instead of 2,137,086 rows, 94.4% of which would be "No").
- gender value Unknown/Invalid (3 rows) stored as Unknown and flagged (DQ12); never imputed.
- Race and gender are taken per patient: the first non-null value by lowest encounter_id; a conflict inside a file is flagged (DQ13).

3.4 Data-quality rules (code: src/etl/quality.py; full table: docs/data_contract.md section 8b)

Rule	Check	Action
DQ01	key columns present, integer, positive	reject
DQ02	duplicate encounter_id in the file	keep first, reject rest
DQ03	admission type, source or disposition id not in the reference tables	reject

Rule	Check	Action
DQ04	value outside the allowed set (readmitted , age , gender , drug status, lab results, change , diabetesMed)	reject
DQ05	numeric ranges (stay 1-14; counts non-negative and below caps)	reject
DQ06	ICD-9 format (short numeric codes are valid: the source dropped leading zeros)	set code to NULL and flag
DQ07	dates missing or not ISO YYYY-MM-DD	reject
DQ08	dates outside 1999-01-01..2008-12-31 (discharge up to 14 days later) or in the future	reject
DQ09	discharge before admission	reject
DQ10	date difference differs from time_in_hospital	reject
DQ11	missing values per column	metric only
DQ12	gender Unknown/Invalid	keep as Unknown , flag
DQ13	conflicting race or gender for one patient in a file	keep first non-null, flag

A file in which more than 20% of rows are rejected fails as a whole and loads nothing; otherwise good rows load and bad rows are written to `data/rejected/` with the rule and reason. On the genuine data every batch is rejected **0 rows**; the rules are demonstrated on a deliberately faulty batch (23 rows rejected, 5 corrected).

3.5 Derived fields

Field	Definition
readmitted_30d	readmitted == '<30'
any_readmission	readmitted is <30 or >30
is_readmission_eligible	false when discharge_disposition_id is 11, 13, 14, 19, 20 or 21 (expired or hospice)
age_order	position of the age bracket, 1-10

4. The readmission rate

The 30-day readmission rate is `readmitted_30d` among eligible encounters divided by the number of **eligible** encounters. Patients who died or entered hospice could not be readmitted, so they leave the denominator (2,423 encounters). On the full data this is $11,314 / 99,343 = \mathbf{0.113888}$ (11.39%). Dividing by all 101,766 encounters would give 0.111177, which understates the rate without raising any error. The rate is per encounter, not per patient.

5. SIMULATED fields (not in the source)

The source has **no dates**. To support trends, date-range queries and incremental loading, two fields are created:

Field	Rule
admission_date	Simulated. Seeded random dates (<code>numpy.random.default_rng(seed=42)</code>) between 1999-01-01 and 2008-12-31 with mild growth and winter seasonality, sorted and assigned in <code>encounter_id</code> order. Re-running produces identical files.

Field	Rule
discharge_date	Simulated. admission_date + time_in_hospital days.

Every dashboard, API response (`meta.dates_simulated = true`), report and document that shows a date says so. Monthly or yearly patterns are illustrative and must never be read as clinical or epidemiological findings.

6. Other limitations of the data

- **No hospital identifier:** hospital-level comparisons are impossible; admission type, admission source, discharge disposition and specialty are compared instead.
- **Missingness:** `medical_specialty` 49.08% and `payer_code` 39.56% missing; category comparisons describe only the recorded subset.
- **Repeated patients:** encounters of one patient are not independent.
- **Dated and observational:** 1999-2008 data; findings are associations, not causes, and do not describe current practice.
- **Coded unknown categories** (for example discharge disposition 18, admission source 17) are kept as unknown, not imputed.

7. Privacy note

The extract is de-identified research data released for public use. The system still treats it as sensitive: `race` and `gender` exist only on the `patients` table, are visible only to administrators and clinical operations staff, never appear in exports, analytics, logs or error messages, and analytics groups with fewer than 11 encounters are flagged `small_n`. Analysts see encounters without `patient_nbr`. Raw data files are not committed to the repository. The system is for education only and must not be used for clinical decisions.